

SPIRIT RIVER, AB, Canada

WMO: 715710

Lat: 55.695N

Lon: 119.230W

Elev: 3331

StdP: 13.01

Time Zone: -7.00 (NAM)

Period: 99-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-20.0	-15.7	-25.8	1.5	-20.0	-20.8	2.0	-15.7	22.3	27.1	20.8	27.7	6.5	310	0.718

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	77.1	59.2	73.6	57.7	70.5	56.4	61.6	72.9	59.6	70.0	57.8	67.4	7.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.4	79.7	64.8	55.4	74.0	62.4	53.7	69.3	61.4	29.1	73.3	27.6	70.4	26.4	67.5	69.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	17.8	16.2	-24.8	83.7	6.2	3.9	-29.2	86.5	-32.9	88.7	-36.4	90.9	-40.9	93.8
			-24.8	65.1	6.2	2.0	-29.3	66.6	-32.9	67.8	-36.4	68.9	-40.9	70.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.8	16.5	19.0	23.1	36.6	47.5	55.3	59.2	56.9	49.2	37.2	23.2	16.2	(6)
	DBStd	19.63	16.92	14.36	14.41	9.51	8.75	6.14	6.14	7.01	8.33	9.67	13.43	15.45	(7)
	HDD50	5716	1039	867	835	410	148	19	5	19	112	410	804	1048	(8)
	HDD65	10349	1504	1287	1300	852	542	294	198	264	478	863	1254	1513	(9)
	CDD50	883	0	0	0	8	72	179	291	234	87	12	0	0	(10)
	CDD65	40	0	0	0	0	1	4	18	14	3	0	0	0	(11)
	CDH74	279	0	0	0	1	8	28	132	90	20	0	0	0	(12)
	CDH80	41	0	0	0	0	0	2	23	14	2	0	0	0	(13)
(14) Wind	WSAvg	9.0	9.9	9.3	8.6	8.7	8.5	8.0	8.3	8.3	9.3	9.4	9.6	10.0	(14)
(15) Precipitation	PrecAvg	18.7	1.3	0.8	1.0	1.0	1.7	2.8	2.9	2.0	1.5	1.3	1.3	1.1	(15)
	PrecMax	26.9	3.6	1.9	1.9	2.0	3.5	5.7	5.2	5.3	4.3	3.7	3.5	4.2	(16)
	PrecMin	12.1	0.6	0.1	0.5	0.3	0.3	0.8	0.9	0.5	0.4	0.3	0.3	0.1	(17)
	PrecStd	3.6	0.7	0.4	0.4	0.4	0.9	1.4	1.3	1.2	0.9	0.7	0.7	0.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.5	46.7	50.7	65.7	75.0	77.4	83.1	81.7	77.1	66.1	51.3	46.4	(19)
		MCWB	37.6	36.4	39.5	46.8	53.0	57.7	62.4	61.1	58.3	50.8	41.3	37.4	(20)
	2%	DB	42.8	42.1	46.0	58.2	69.0	73.4	77.5	76.0	70.5	58.6	46.3	42.3	(21)
		MCWB	35.1	34.7	36.4	43.7	50.4	55.5	60.3	60.2	56.1	47.3	39.0	35.3	(22)
	5%	DB	39.2	39.2	42.9	53.4	64.6	70.1	73.6	71.7	65.6	53.5	42.1	39.0	(23)
		MCWB	34.2	32.6	34.7	40.8	48.5	54.7	58.9	58.2	53.7	44.7	36.1	34.0	(24)
	10%	DB	36.2	36.2	39.6	49.5	61.1	66.6	70.2	68.1	61.2	49.6	38.6	35.6	(25)
		MCWB	32.1	30.9	32.5	38.6	46.8	53.0	57.4	56.3	51.7	42.1	33.9	31.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.0	38.1	40.8	47.9	54.9	61.2	64.8	64.3	60.6	52.3	43.0	39.0	(27)
		MCDB	45.2	44.1	49.5	63.1	68.8	71.4	76.3	75.6	72.0	64.1	48.8	44.1	(28)
	2%	WB	36.5	35.4	37.6	44.4	51.9	58.0	62.3	61.1	57.1	48.4	39.7	36.3	(29)
		MCDB	41.8	41.1	44.6	57.2	64.6	69.2	74.3	72.0	67.2	56.7	45.0	41.0	(30)
	5%	WB	34.5	33.1	35.1	41.6	49.9	55.9	60.0	59.3	54.5	45.5	37.3	34.1	(31)
		MCDB	38.8	38.5	41.9	51.4	62.0	66.9	70.6	69.2	63.8	53.0	41.6	38.8	(32)
	10%	WB	32.4	31.0	33.1	39.2	47.9	54.1	58.1	57.4	52.0	42.7	34.5	31.6	(33)
		MCDB	35.8	35.8	39.1	48.4	58.5	64.2	68.0	65.9	59.6	48.8	38.1	35.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.6	13.5	15.5	16.1	17.2	16.6	16.6	16.3	15.2	11.8	10.8	11.9	(35)
		MCDBR	12.2	13.9	17.2	20.2	21.3	20.6	21.0	20.6	20.6	17.2	12.1	11.5	(36)
	5% WB	MCWBR	8.9	8.8	10.5	11.1	10.3	9.6	10.0	9.7	9.9	10.2	8.3	8.6	(37)
		MCWBR	12.4	13.7	16.6	19.1	19.8	18.4	19.2	19.1	19.8	16.1	12.6	11.9	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.229	0.237	0.285	0.302	0.325	0.327	0.327	0.327	0.292	0.271	0.230	0.214	(40)
		taud	2.311	2.373	2.304	2.435	2.437	2.443	2.471	2.455	2.568	2.578	2.429	2.265	(41)
	Ebn at Noon	Ebn at Noon	233	277	282	294	291	291	289	282	281	263	235	210	(42)
		Edh at Noon	16	22	30	31	33	33	32	30	23	18	14	13	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	228	527	998	1491	1779	1936	1883	1529	1046	558	250	161	(44)
	RadStd	RadStd	24	41	79	82	120	152	117	135	92	63	22	14	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+256	N/A
(47) Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page